



Line Pressure Surveillance & Impact: IAM Approach

August 28th, 2018



Team



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Agenda



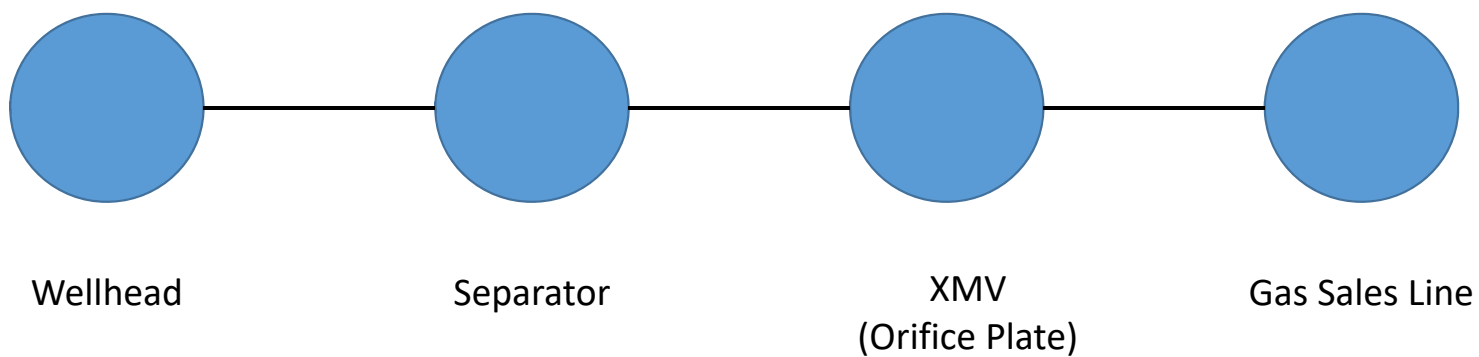
- Executive Summary
- Process Flow
- Value Proposition
- Historical Line Pressure Impact
 - Volume & Revenue Impact
- Well Case Studies
- Path Forward

Executive Summary

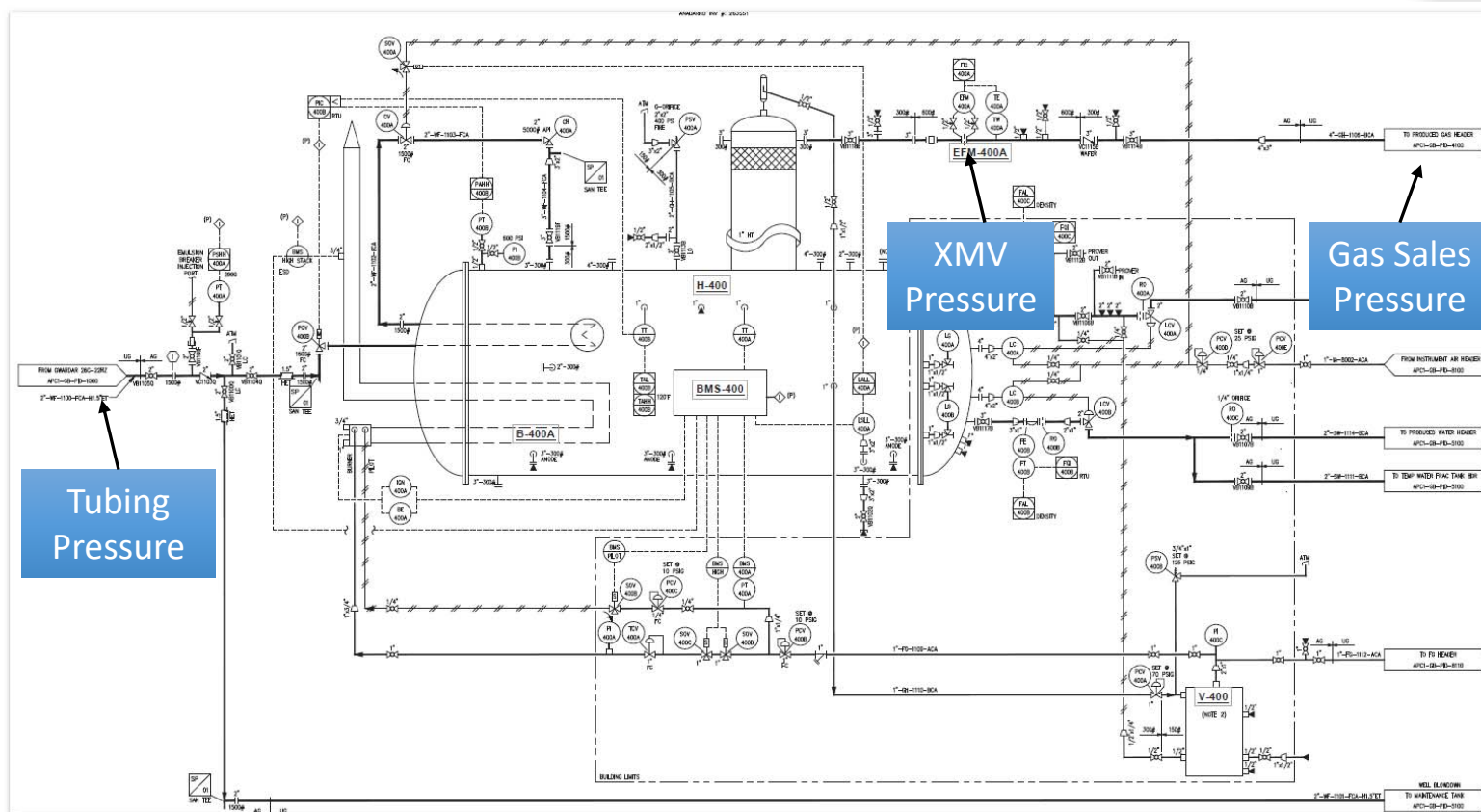


- Lower line pressure has always been desired by production
 - Assumed high Impact during summer due to compressors
 - Pressure equals money:
 - Pressure Increase = Production Decrease & Deferred Revenue
 - Pressure Decrease = Expenditure & Capital Spend
 - Need to find the equilibrium point between the two
- Historically, no good way of quantifying the problem
 - Manual process: Too many wells & variables
- Today: We have leveraged Digital Oil Field technology to quantify line pressure impacts automatically and showcase value
- Potential value estimated for APC:
 - +/-2,000 bpd and +/- \$11MM @ \$60/bbl

Process Flow



Process Flow



Value Proposition



Elevating the IOC & Field Operations

Real time oil / gas opportunity loss analysis by well



Enhanced daily & weekly IOC reporting

Identifying high pressure 'quadrants' of the field



Turn compression on earlier & enhanced KMOG/WES alignment

High backpressure inside the production facility



Piping and control valve modifications to reduce wellhead back-pressure

Identifying wells that are flowing in subsonic flow that are more susceptible to line pressure increases



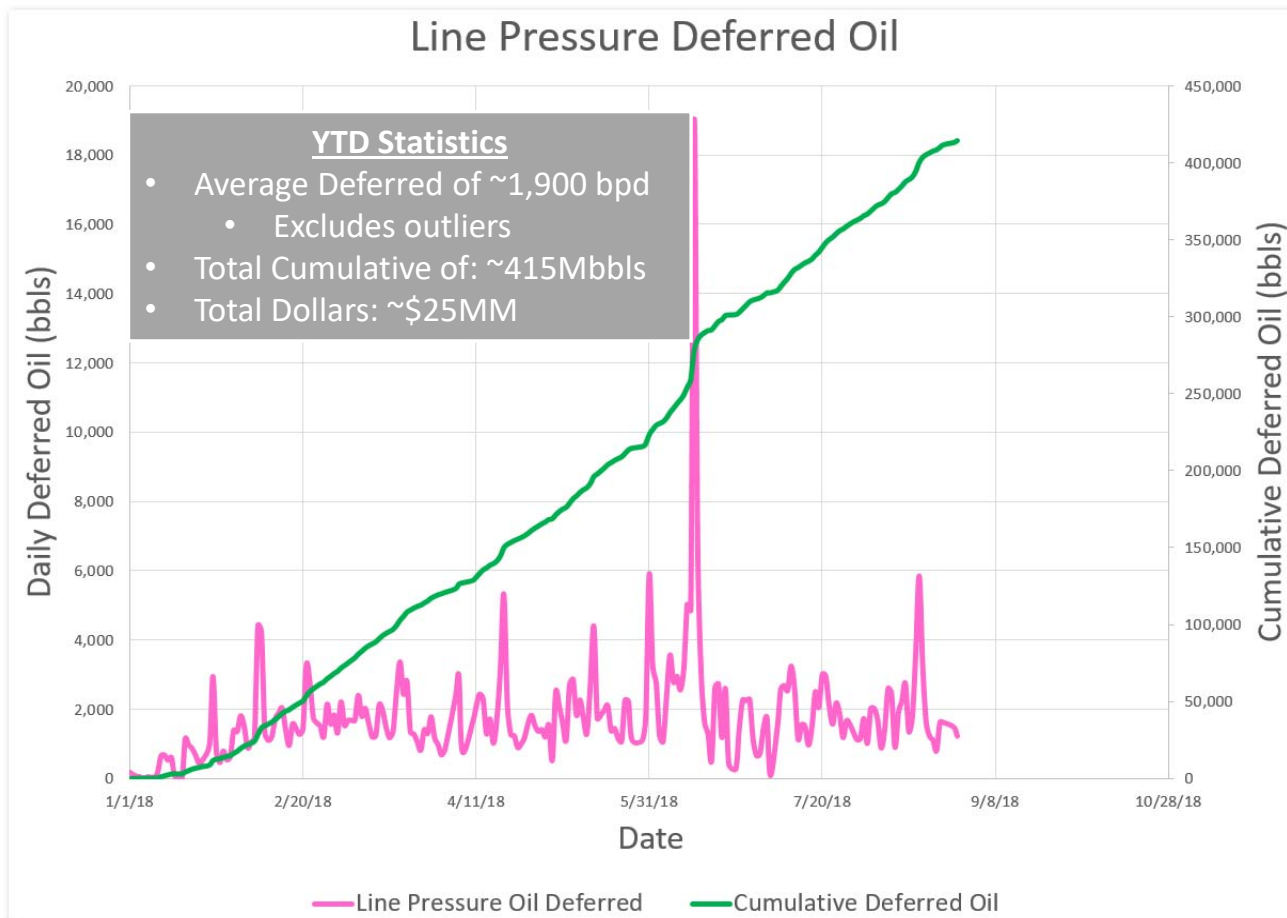
Tune wellhead telemetry to increase production

Workflow programmed to pull historical line pressure averages by production battery



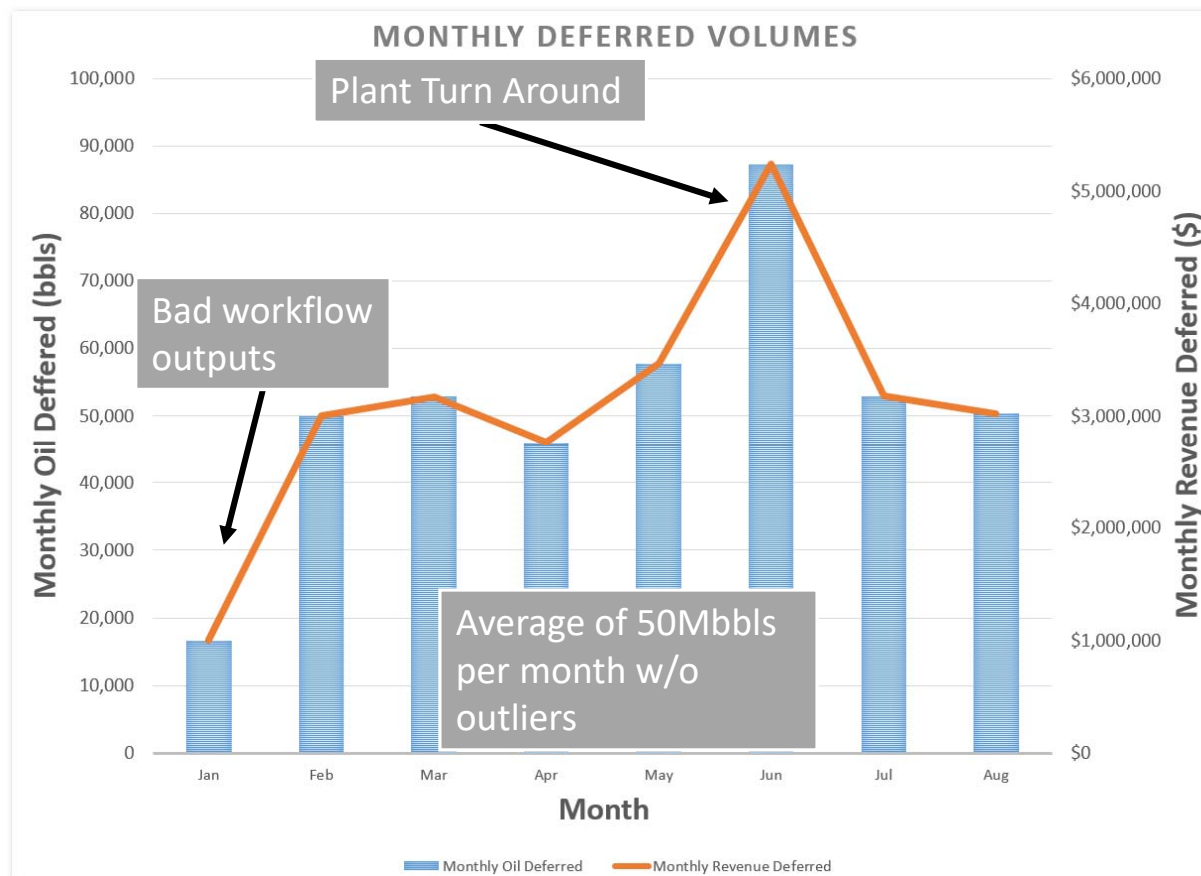
Data support in KMOG/WES contract discussions

Economics – Daily Deferred Volumes



8/28/2018

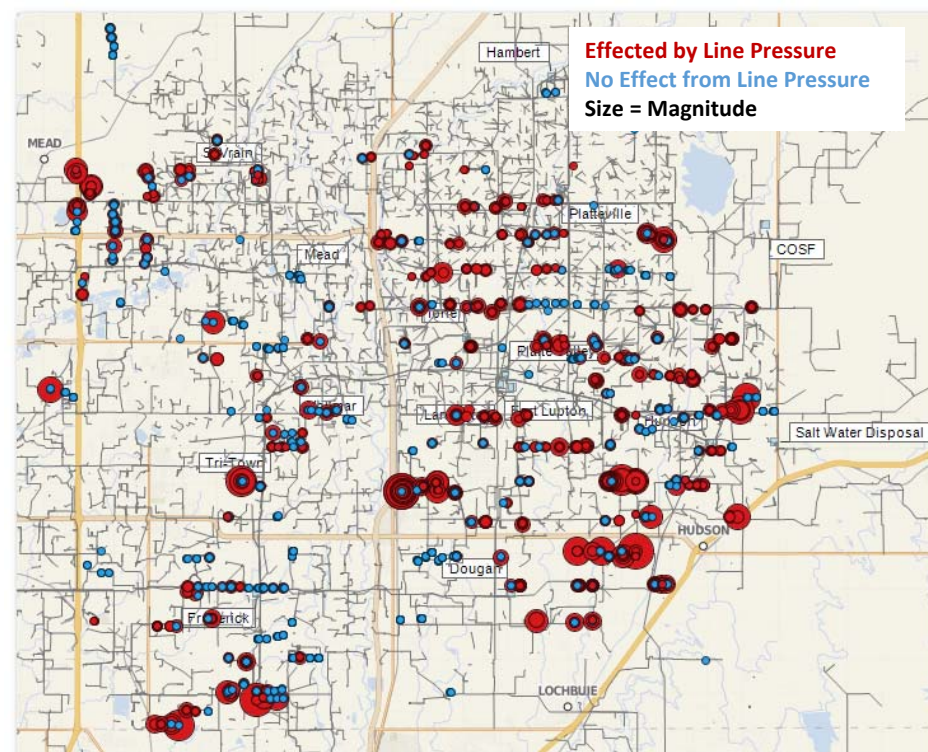
Economics – Monthly Deferred Volumes



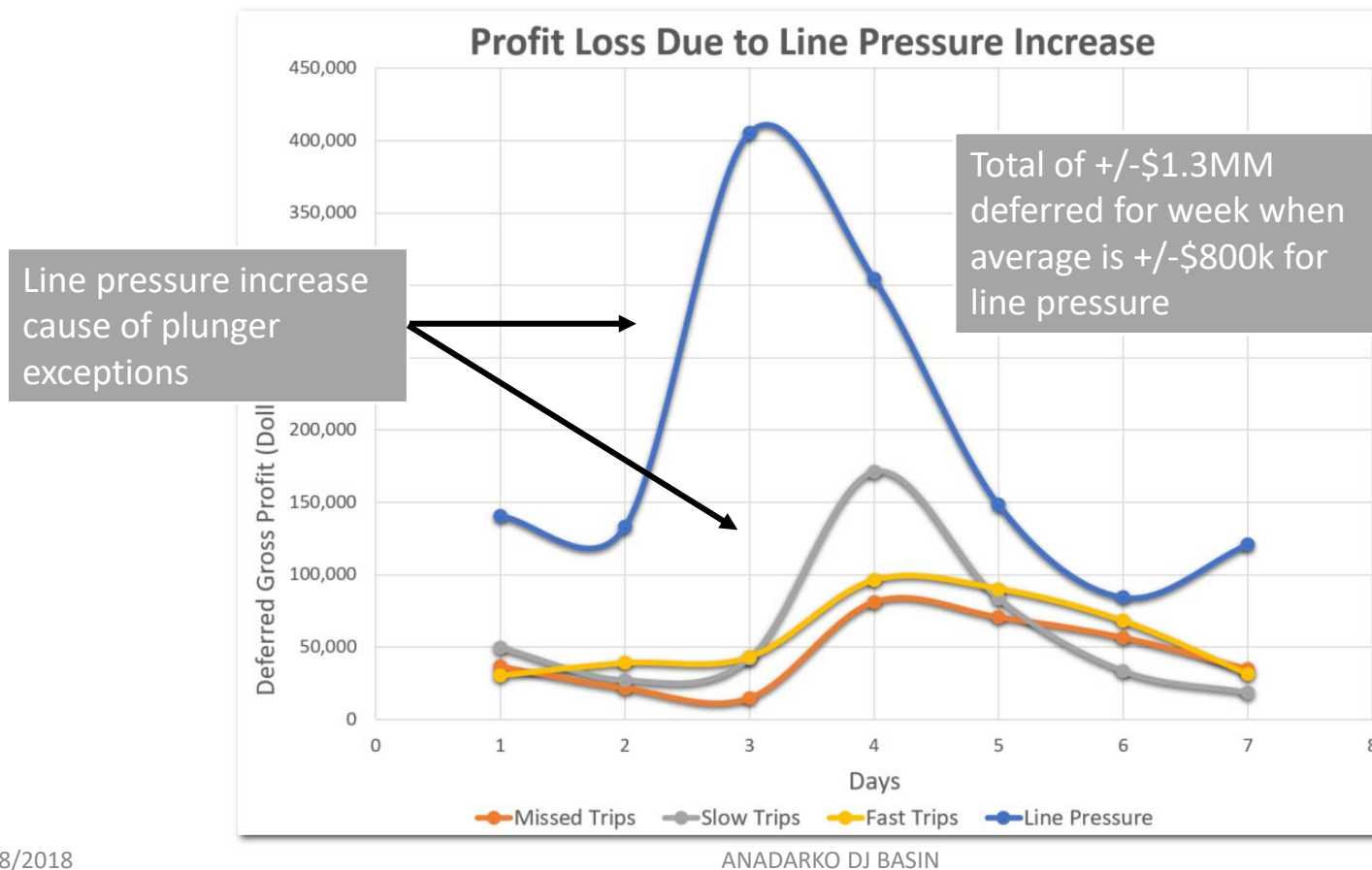
Economics – Map of Deferment



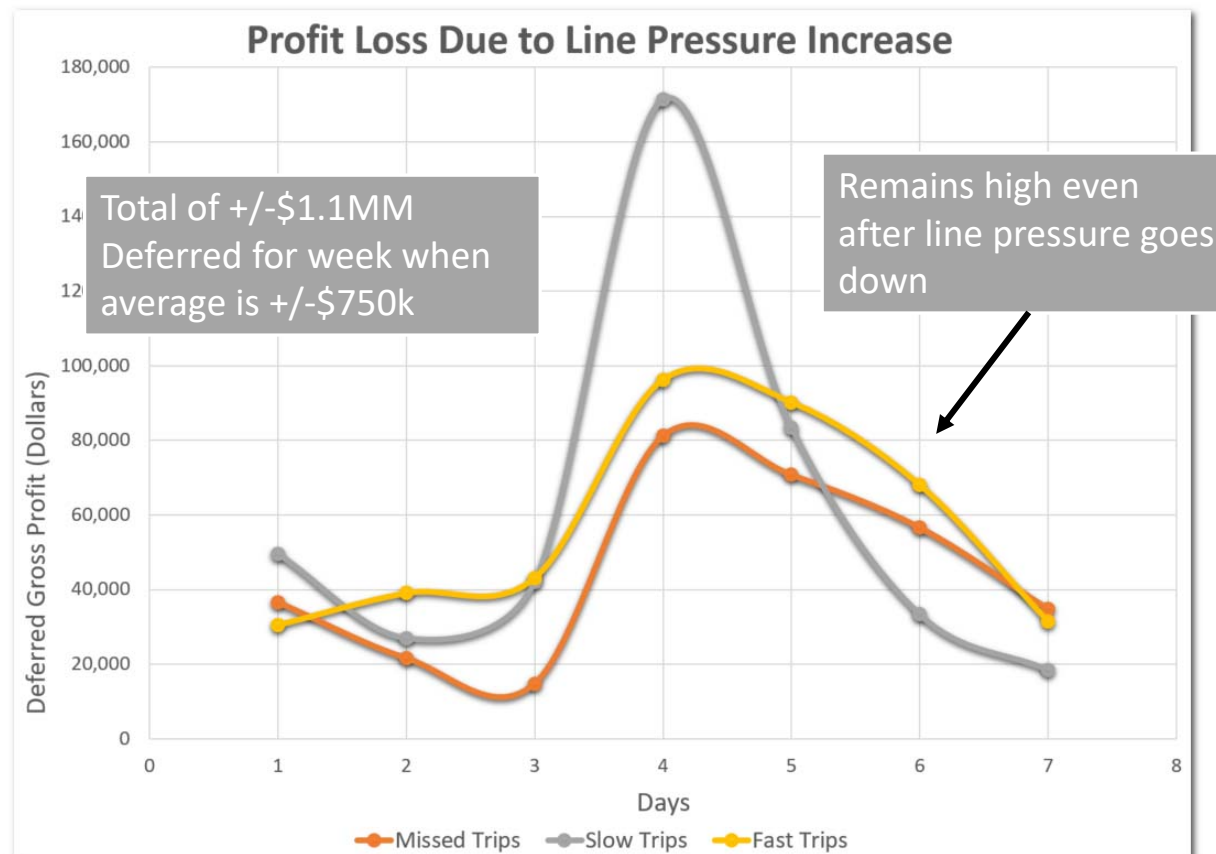
- Time of Map 8/15/18 - 8/22/18
 - Size based on 7 days of deferment
- Distribution spread across field
 - Mixture of wedge & base wells



Economics – Line Pressure Impact



Economics – Line Pressure Impact



XMV Pressure Exception



Field Oil Differed (BBLs): 12,832
Field Sales Differed (\$): \$898,274

XMV Pressure Exception

Gas Sales Exception XMV to Line Delta Exception Summary

Pressure Bins

Pressure Deltas

Daily Deferred Volumes & Sales

Weekly Deferred Volumes & Sales

Well Name	Tubing Pressure Bin	XMV Pressure Bin	Gas Sales Pressure Bin	XMV to Sales Press Delta	8/1/2018 4:00:00 AM	8/2/2018 4:00:00 AM	8/3/2018 4:00:00 AM	8/4/2018 4:00:00 AM	8/5/2018 4:00:00 AM	8/6/2018 4:00:00 AM	8/7/2018 4:00:00 AM	Totals
on Ranch 33C-27HZ	500 psi - 550 psi	150 psi - 200 psi	100 psi - 150 psi	45	0	101	0	0	0	0	0	101
ein 3C-28HZ	300 psi - 350 psi	100 psi - 150 psi	100 psi - 150 psi	22	\$0	\$7,044	\$0	\$0	\$0	\$0	\$0	\$7,044
Jaguar 3C-14HZ	850 psi - 900 psi	100 psi - 150 psi	150 psi - 200 psi	-42	13	14	13	14	14	10	21	99
Weichel 30N-23HZ	900 psi - 950 psi	150 psi - 200 psi	150 psi - 200 psi	0	\$937	\$950	\$889	\$948	\$985	\$728	\$1,498	\$6,935
Bbkb 16N-A35HZ	950 psi - 1000 psi	150 psi - 200 psi	150 psi - 200 psi	1	0	0	0	0	0	0	88	88
Ss Minnow 23N-14HZ	750 psi - 800 psi	150 psi - 200 psi	150 psi - 200 psi	1	\$0	\$0	\$0	\$0	\$0	\$0	\$6,160	\$6,160
Small Eyed 14C-35HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	23	0	15	9	31	17	8	8	88
Wardell 28N-20HZ	150 psi - 200 psi	100 psi - 150 psi	100 psi - 150 psi	11	\$0	\$1,033	\$602	\$2,181	\$1,179	\$539	\$594	\$6,128
Underhill State 26C-17HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	9	0	0	0	30	9	0	48	87
Dove 8C-13HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	10	\$0	\$0	\$0	\$2,133	\$653	\$0	\$3,327	\$6,113
Paragon 40C-31HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	10	48	0	0	0	0	0	39	86
Melbon 19C-8HZ	100 psi - 150 psi	100 psi - 150 psi	0 psi - 100 psi	15	\$3,335	\$0	\$0	\$0	\$0	\$0	\$2,720	\$6,055
Highway 160 37N-2HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	4	0	0	0	84	0	0	0	84
Troutt 14N-33Hz	200 psi - 250 psi	150 psi - 200 psi	100 psi - 150 psi	43	\$0	\$0	\$0	\$5,904	\$0	\$0	\$0	\$5,904
Cream State 28N-21HZ	150 psi - 200 psi	100 psi - 150 psi	100 psi - 150 psi	40	10	12	12	15	8	12	11	80
					\$671	\$840	\$869	\$1,017	\$586	\$847	\$797	\$5,626
					0	3	17	20	20	20	20	80
					\$0	\$203	\$1,211	\$1,378	\$1,378	\$1,378	\$1,378	\$5,587
					5	0	20	0	0	0	0	72
					\$329	\$0	\$1,376	\$0	\$0	\$0	\$0	\$5,051
					0	24	24	0	0	0	24	72
					\$0	\$1,669	\$1,683	\$0	\$0	\$0	\$1,672	\$5,023
					17	15	12	8	8	0	12	71
					\$1,156	\$1,079	\$838	\$535	\$531	\$0	\$863	\$5,002
					0	0	4	27	0	22	18	71
					\$0	\$0	\$249	\$1,912	\$0	\$1,527	\$1,266	\$4,954
					12	6	2	15	19	0	17	70
					\$826	\$393	\$145	\$1,038	\$1,309	\$0	\$1,208	\$4,918
					15	13	12	5	3	9	12	70
					\$1,070	\$930	\$861	\$352	\$226	\$641	\$820	\$4,901

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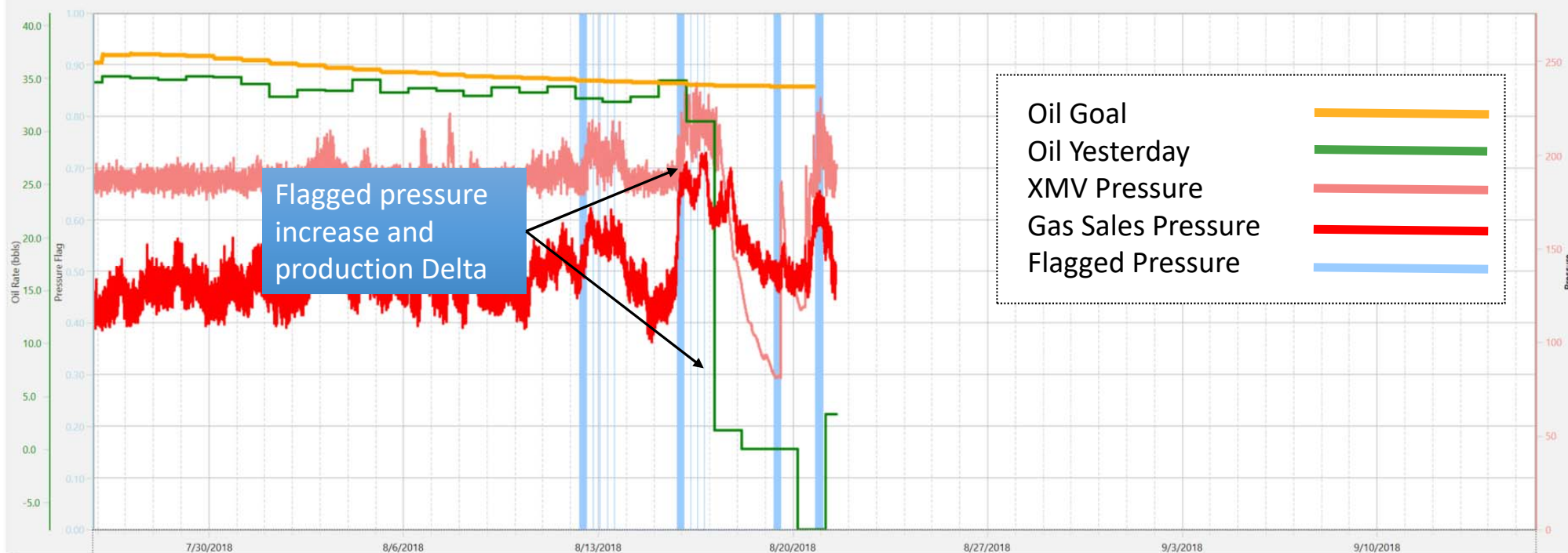
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Flagged Pressure Exception



Hunziker 28N-28HZ Pressure Deferment Plot



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XMV to Gas Sales Line Delta



Field Oil Differed (BBLS):

12,832

Field Sales Differed (\$):

\$898,274

XMV to Gas Sales Line Delta Exception

Gas Sales Exception

Line Pressure Summary

Exception Summary

Well Name	Tubing	XMV Pressure	Gas Sales Pressure Bin	XMV to Sales Press Delta	8/1/2018 4:00:00 AM	8/2/2018 4:00:00 AM	8/3/2018 4:00:00 AM	8/4/2018 4:00:00 AM	8/5/2018 4:00:00 AM	8/6/2018 4:00:00 AM	8/7/2018 4:00:00 AM	Totals
Powers 1C-23HZ			0 psi - 100 psi	126	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 1N-12HZ			100 psi - 150 psi	95	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 26C-12HZ		600 psi - 650 psi	200 psi - 250 psi	100 psi - 150 psi	95	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 26N-25HZ		500 psi - 550 psi	200 psi - 250 psi	100 psi - 150 psi	94	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 1C-25HZ		550 psi - 600 psi	200 psi - 250 psi	100 psi - 150 psi	93	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 27N-25HZ		500 psi - 550 psi	200 psi - 250 psi	100 psi - 150 psi	89	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 27C-12HZ		600 psi - 650 psi	200 psi - 250 psi	100 psi - 150 psi	89	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 26C-12HZ		350 psi - 400 psi	200 psi - 250 psi	100 psi - 150 psi	87	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 16C-25HZ		300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	87	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 2C-12HZ		250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	86	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 15C-25HZ		300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	86	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 15N-25HZ		250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 27N-12HZ		300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 16N-25HZ		250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 37N-25HZXX		200 psi - 250 psi	200 psi - 250 psi	100 psi - 150 psi	84	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0

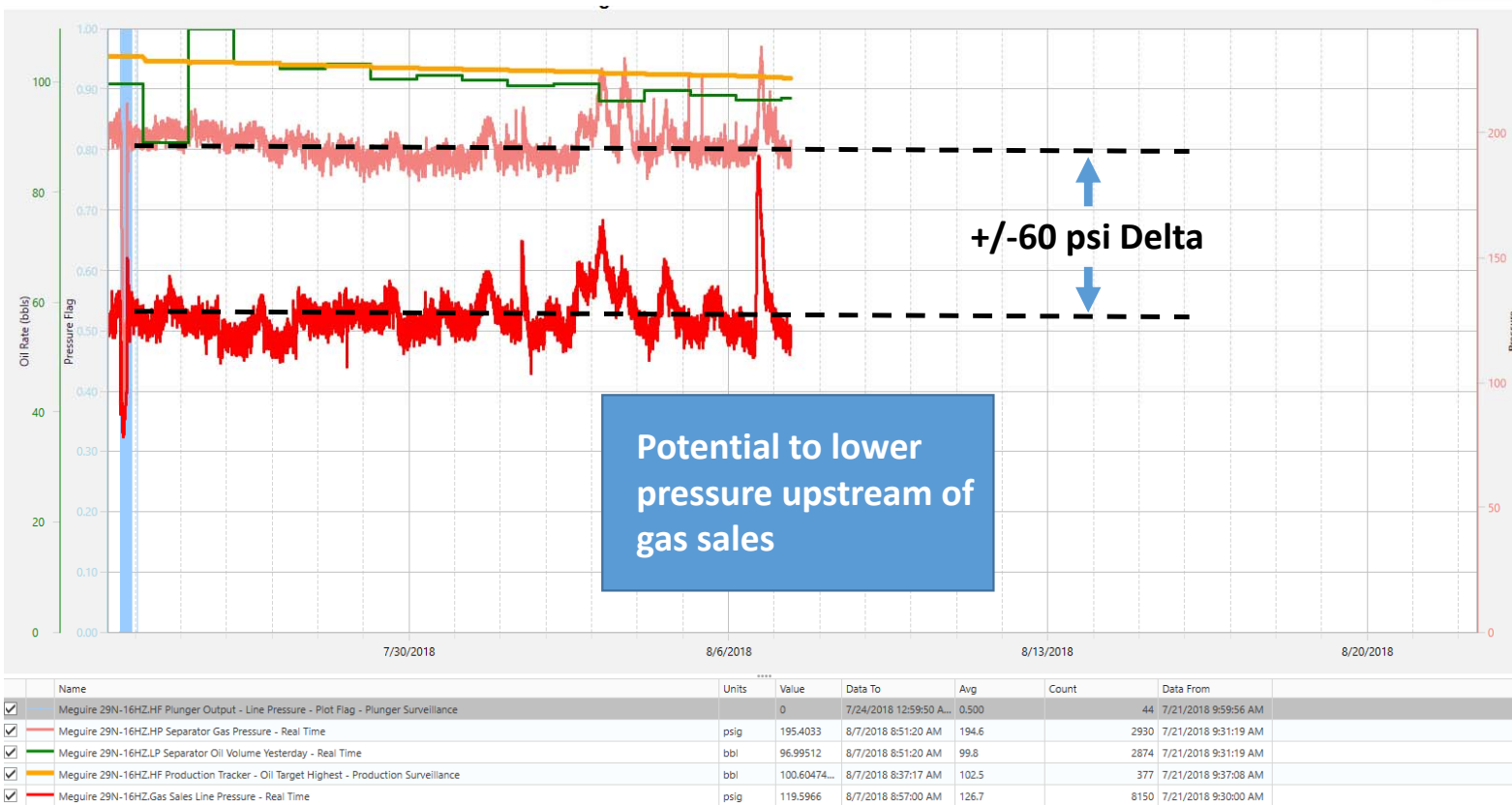
Sorted by
pressure delta

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XMV to Gas Sales Line Delta



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Path Forward



- Generate routine reports for management
 - Quantified data to aid discussions with third parties
- Capture potential volume impacts in budget process
- Understand & capture contract violations automatically
 - Exceed contract thresholds
- Multi-disciplinary team optimization workshop
 - What can we do to lower line pressure?

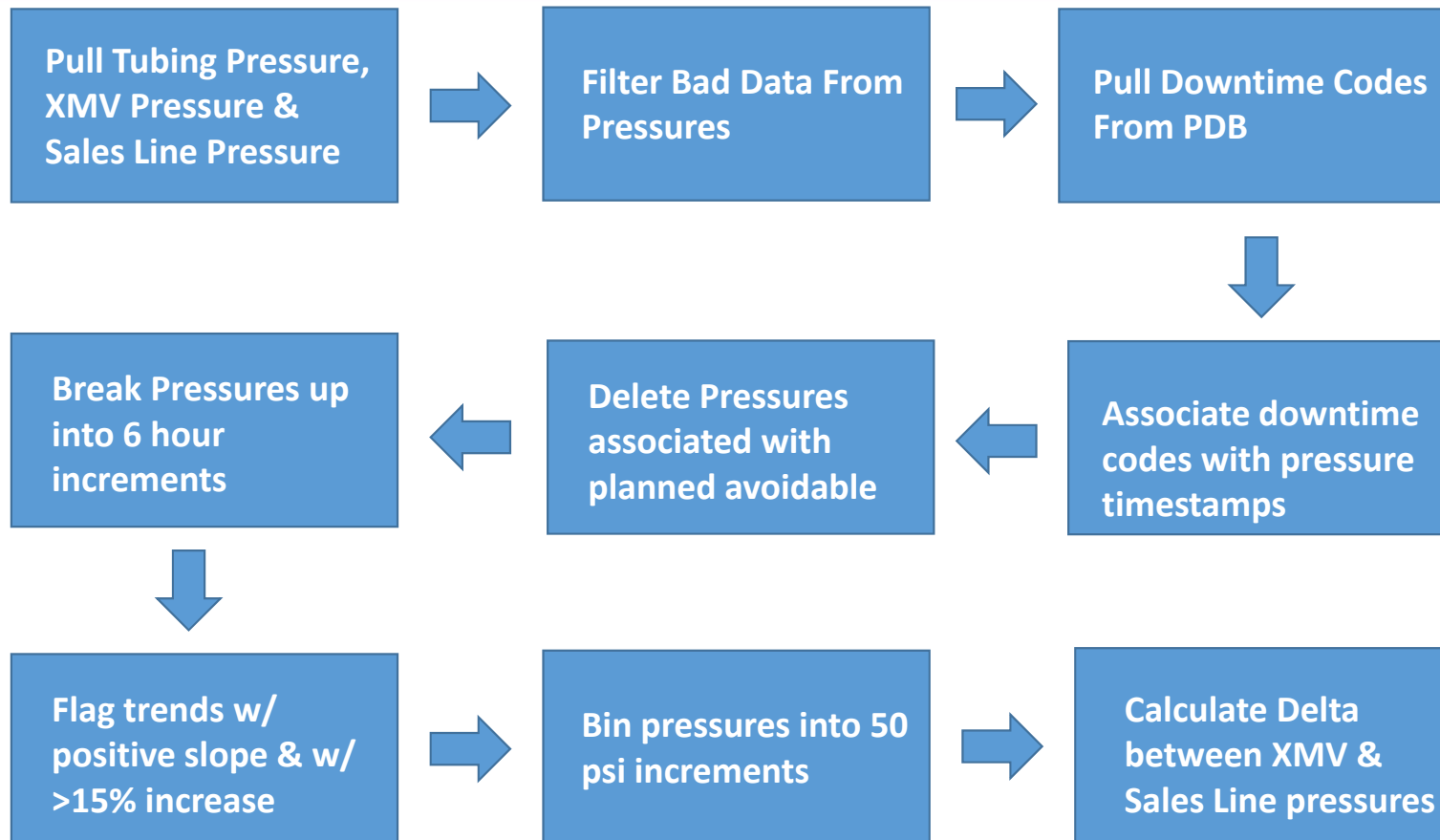
Questions



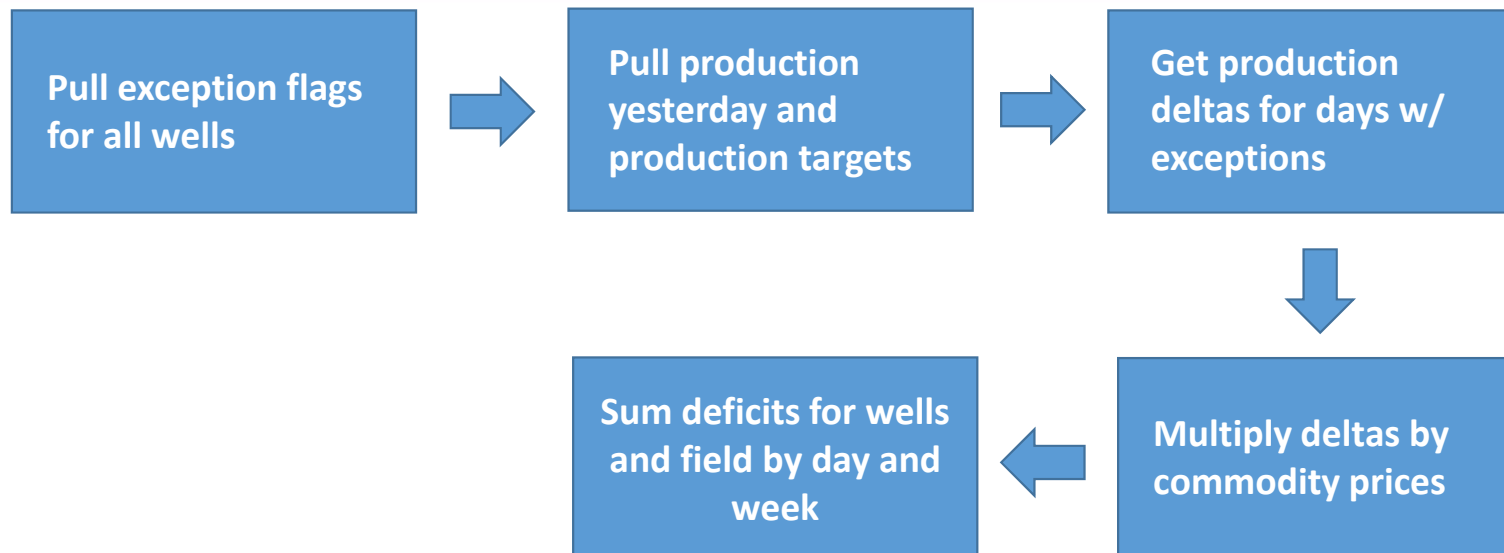
Back up



Workflow 1 Steps



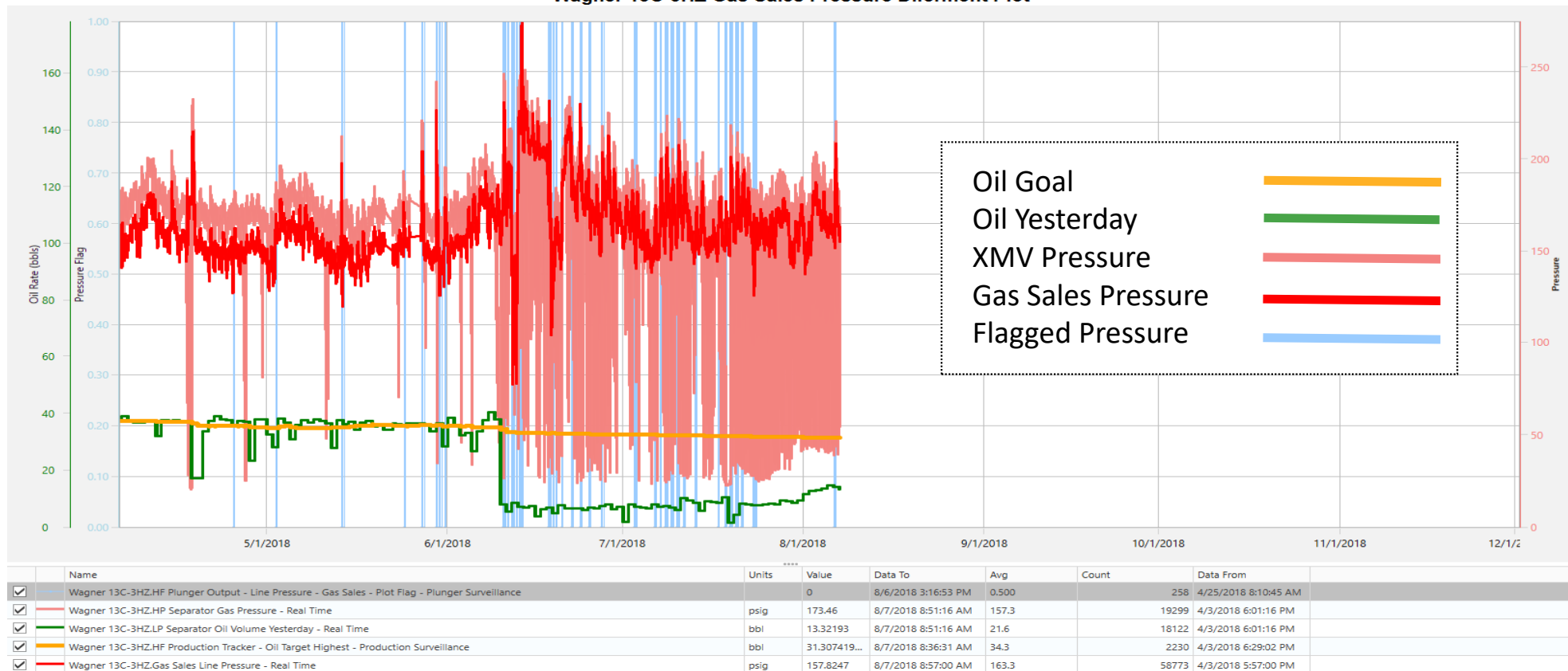
Workflow 2 Steps



Flagged Pressure Exception



Wagner 13C-3HZ Gas Sales Pressure Diferment Plot



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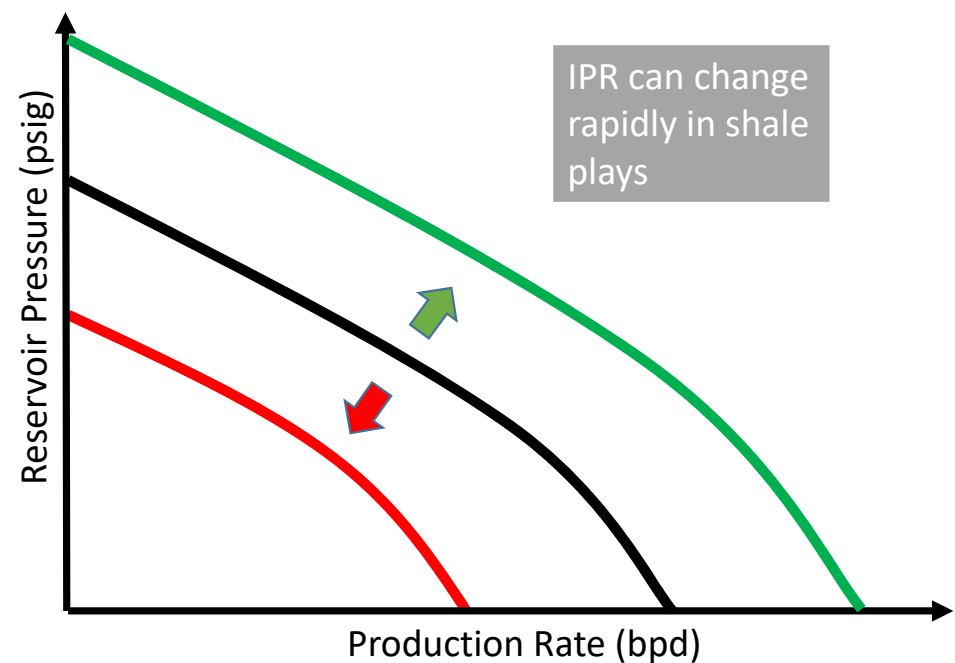
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Inflow Performance Relationship



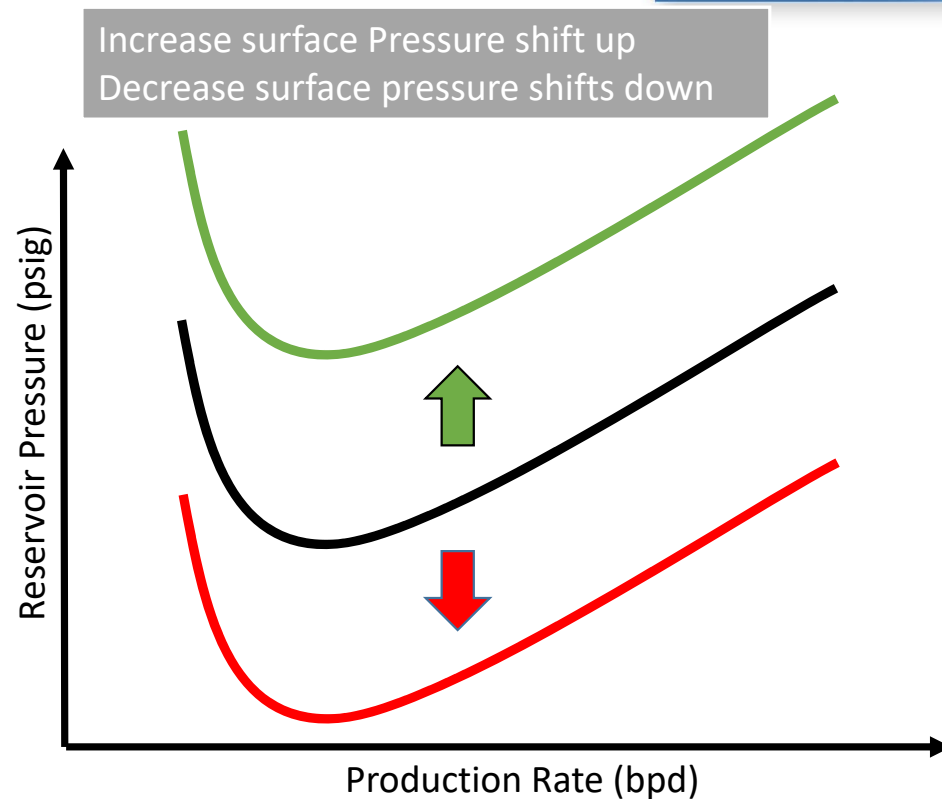
- Relationship Between:
 - Reservoir Pressure
 - Flowing Bottom Hole Pressure
 - Production Rate
- Stable in conventional plays
 - Sand Stone, etc.
- Dynamic in unconventional plays
 - Shale, etc.



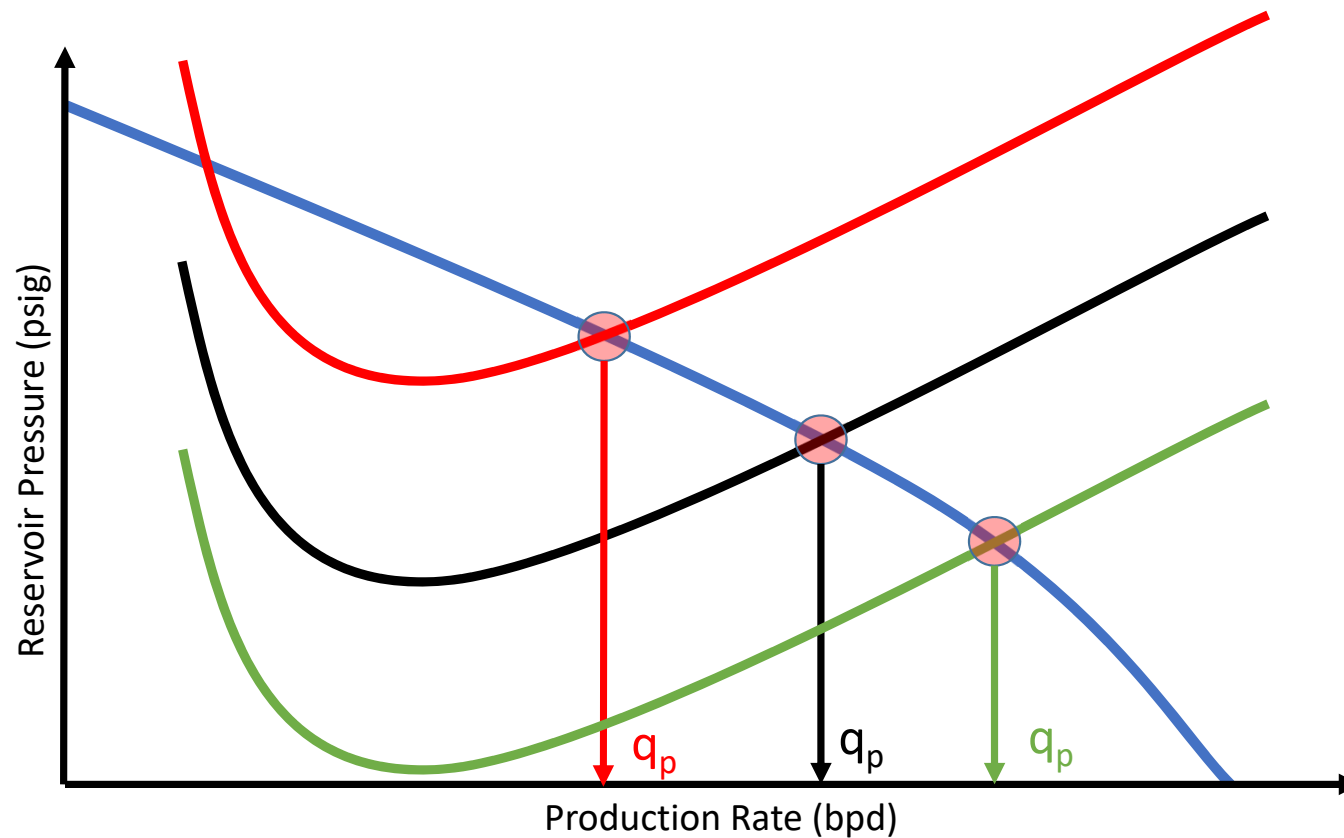
Vertical Lift Performance Relationship



- VLP: Relationship between bottom hole & surface
 - How production flows from bottom hole to surface
- Dependent on:
 - Well Depth
 - Tubing Size
 - Water Cut
 - GOR
 - PVT Properties
 - **Surface Pressure**



Combining IPR & VLP



Intersection is operating point.

Changes in surface pressure change production rates with all other variables held constant

Compressible Flow

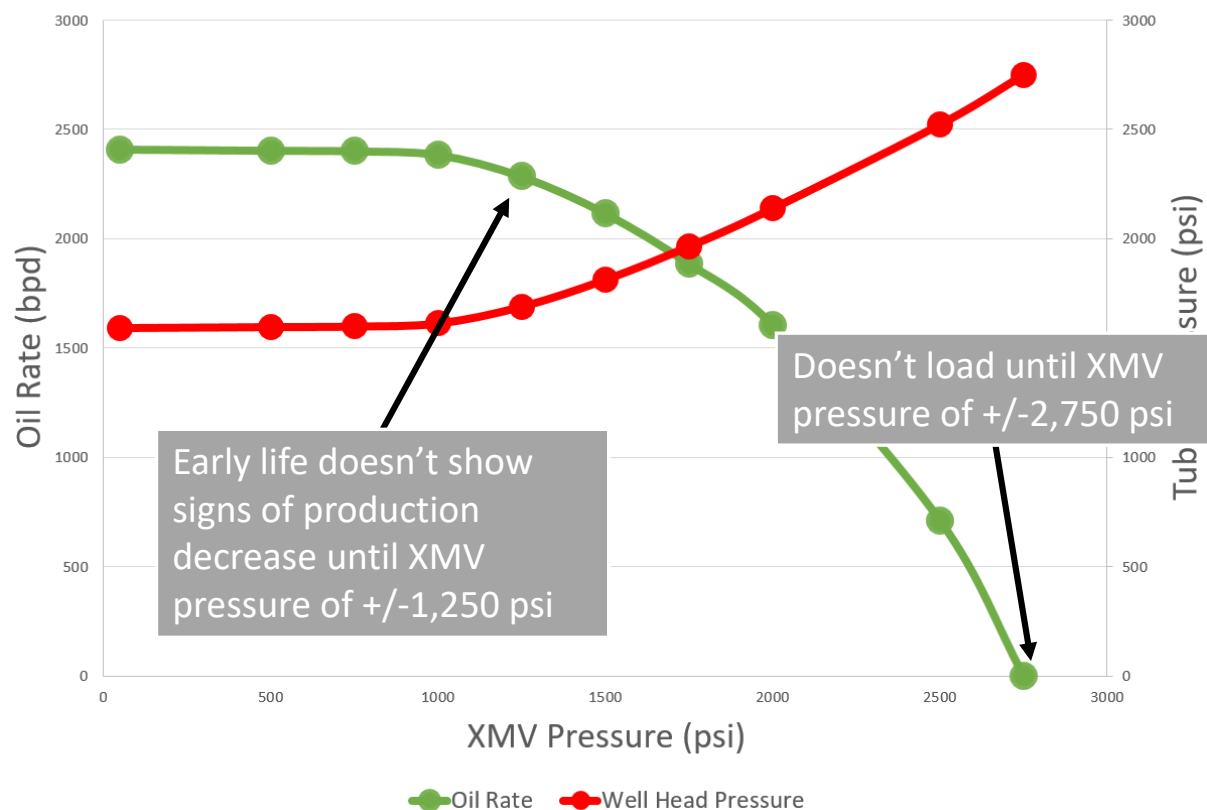


- Gas at subsonic flow acts like an incompressible fluid
 - If line pressure goes up, well head pressure goes up
- Gas at supersonic flow acts like a compressible fluid
 - If line pressure goes up, well head pressure isn't effected

XMV Pressure Sensitivity Example 1



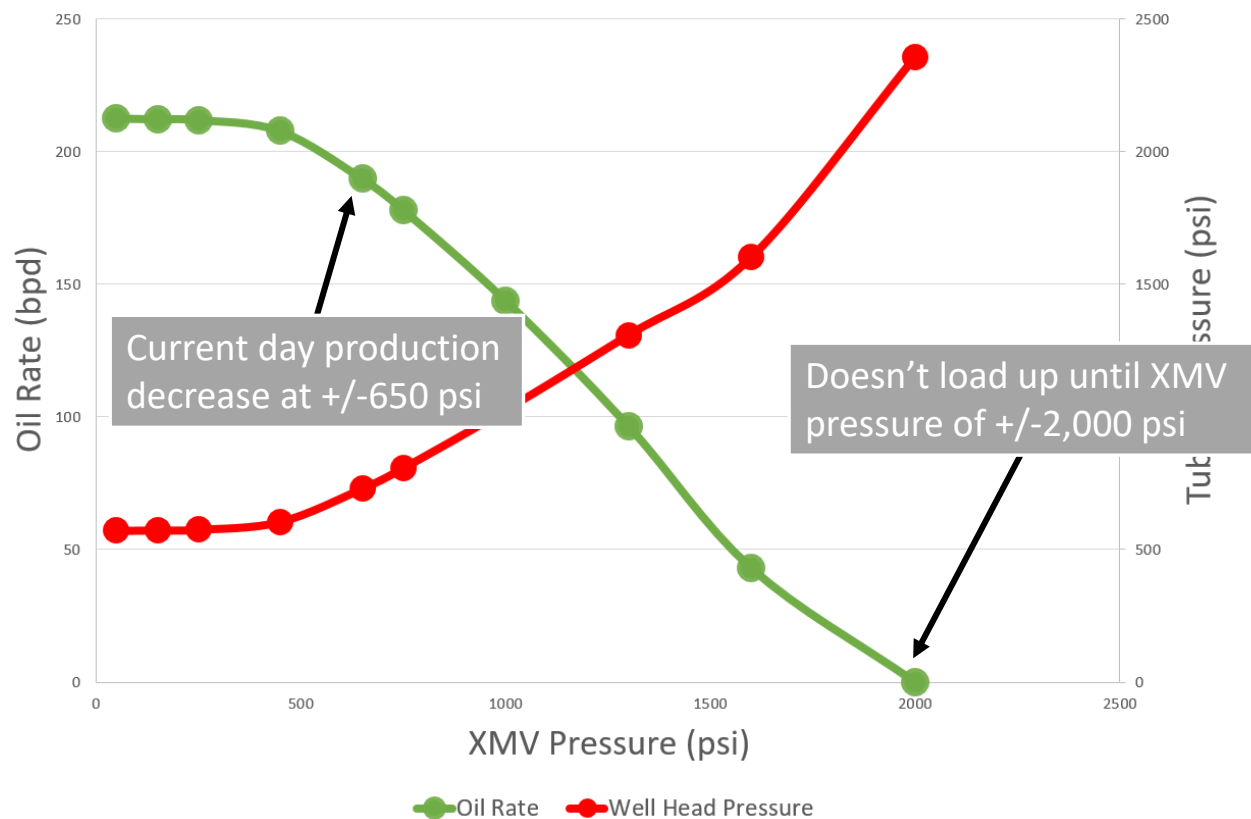
JZM 1C-3HZ XMV Pressure Sensitivity Plot



XMV Pressure Sensitivity Plot Example 2



JZM 1C-3HZ XMV Pressure Sensitivity Plot



Economics – Line Pressure Impact



Field Oil Deferred (BBLs):	43,822							
Field Sales Deferred (\$):	\$3,067,573							
Exception	8/14/2018 4:00:00 AM	8/15/2018 4:00:00 AM	8/16/2018 4:00:00 AM	8/17/2018 4:00:00 AM	8/18/2018 4:00:00 AM	8/19/2018 4:00:00 AM	8/20/2018 4:00:00 AM	Totals
Missed Oil Target	3,354 \$234,785	3,993 \$279,520	5,498 \$384,841	11,094 \$776,600	9,311 \$651,788	6,085 \$425,924	4,487 \$314,115	43,822 \$3,067,573
Missed Trips	610 \$42,701	360 \$25,172	246 \$17,204	1,354 \$94,806	1,180 \$82,633	944 \$66,070	580 \$40,613	5,274 \$369,198
Slow Trips	827 \$57,889	450 \$31,499	701 \$49,097	2,856 \$199,910	1,392 \$97,428	556 \$38,908	308 \$21,571	7,090 \$496,302
Fast Trips	509 \$35,598	652 \$45,626	719 \$50,317	1,603 \$112,201	1,503 \$105,211	1,135 \$79,421	526 \$36,797	6,645 \$465,172
No Flow	478 \$33,452	324 \$22,668	541 \$37,871	670 \$46,869	1,148 \$80,348	809 \$56,652	823 \$57,611	4,792 \$335,470
XMV Pressure	1,910 \$133,699	1,799 \$125,898	3,618 \$253,226	4,590 \$321,315	2,580 \$180,577	1,406 \$98,439	1,230 \$86,101	17,132 \$1,199,255
Gas Sales Pressure	2,343 \$164,014	2,218 \$155,239	6,759 \$473,150	5,078 \$355,454	2,473 \$173,095	965 \$67,565	2,015 \$141,060	21,851 \$1,529,578